NIHAL ERNEST

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EDUCATION

University of California, Santa Cruz (UCSC)

- **M.S. Computer Science** GPA: 3.85 *Sep. 2024 – Mar. 2026*
- **B.S. (Honors) Computer Science** GPA: 3.62 *Jul. 2020 – Jun. 2024*

EXPERIENCE

Stealth Startup *Jun. 2025 – Jun. 2026*

Full Stack/AI Engineer

- Architected a full-stack platform automating real estate investment workflows, with a **Python/Flask** REST API, **SQLite** data layer, and an LLM agent (**OpenClaw** + **Claude**) exposed via Slack through **MCP** on **AWS**. Directly informed **multi-million dollar real estate decisions**.
- Leveraged **SQLite** caching and parallel processing across data ingestion pipelines to reduce an 8-hour process to **under 20 minutes (~24x speedup)**.
- Designed **RESTful** API routes aggregating multi-source geospatial and third-party real estate data, with authentication, observability, and a **git-versioned prompt manager** supporting controlled LLM iteration.

Legal Forensics AI *Jun. 2024 – Sep. 2024*

Software Engineering Intern

- Owned end-to-end development of a full-stack web app enabling legal researchers to query **millions of case documents** in natural language, with an **Angular/TypeScript** frontend and **Python** backend powered by **Llama 3** on **AWS Bedrock**.
- Replaced hours of manual keyword filtering with **LLM**-driven semantic search by integrating the Court Listener **REST API**, **surfacing relevant precedents in seconds**.

Xen.AI *Jun. 2023 – Jun. 2024*

Software Engineering Intern

- Built multi-tenant **Angular/TypeScript** frontends and **Python** backend function apps on **Microsoft Azure** for **enterprise pilots** in banking and healthcare.
- Integrated **OpenAI GPT-4** and **SQL** databases via **REST** endpoints, implementing **Azure** authentication and search services with **tenant-isolated access controls** to meet banking compliance requirements.

RESEARCH

Sloth Lab, UC Santa Cruz *Sep. 2025 – Present*

Graduate Researcher

- Co-authored **Selma**, a multi-agent simulation of opinion dynamics where agents with distinct personalities and knowledge bases debate, with **HDBSCAN** speaker clustering and satisfaction-based conversation exits.
- Built the **LLM natural-language generation pipeline**, grounding each generated line in live agent state.
- Three papers published at **AAAI AIIDE 2026**: *Selma* (**oral**), *On the Reproducibility of Technical Game AI Research* (poster), *Opinionated Agents in Minecraft* (case study).

PROJECTS

Fishing Platform *Feb. 2026 – Present*

- Built a full-stack catch-logging platform (**React** + **Python** REST API), **actively used** by anglers to log catch data into a database-backed metadata layer for historical tracking and cross-user comparison.
- Developed a **recommendation engine** suggesting fishing plans by time, location, gear, and target species from catch history, iterated through **user feedback**.

Formula Forecast *Sep. 2025 – Present*

- Architected a full-stack F1 prediction app (**React** + **FastAPI** + **PyTorch**): a **5-seed ensemble** predicting each driver's deviation from qualifying order, with a **Huber loss** so DNFs and crashes don't dominate.
- Engineered 15 features per driver-race from **FastF1** telemetry (2018–2025) and evaluated against a **fully held-out 2026 season**, beating qualifying order on **exact-position accuracy (38% vs 36%)**.

TECHNICAL SKILLS

Languages	Python, JavaScript, TypeScript, Java, C/C++, HTML/CSS, SQL (PostgreSQL, MySQL, SQLite)
Frameworks	React, Angular, React Native, FastAPI, Flask, PyTorch
Cloud & Tools	AWS, Azure, Git, GitHub Actions (CI/CD), Linux, REST APIs, OpenAI, Llama, Supabase, Selenium